

Features

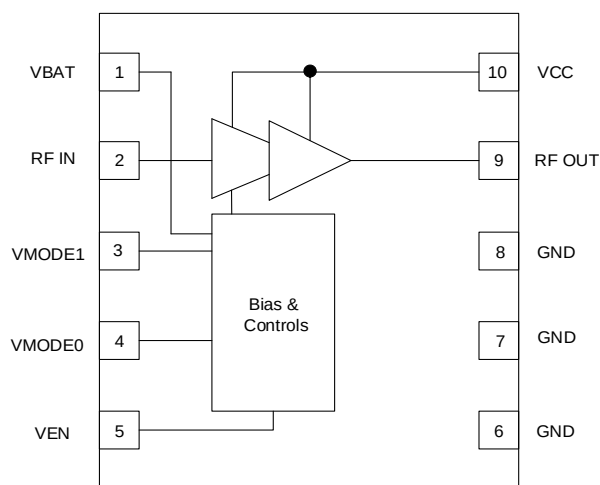
- Optimized for 800–960 MHz operation
- 4.0V nominal operating voltage
- Output power greater than 30dBm
- High PAE at maximum output power
- High/Low power mode control
- Low leakage current
- Single-ended 50 Ω input and output ports
- Fully on-chip matching circuitry

Applications

- Wi-SUN
- LoRa
- RFID
- Industrial IOT
- Connected Homes
- Automatic meter readers

Product Highlights

- Small package size
- Support low voltage vbat
- High Efficiency



Block Diagram

Description

The Geo-Chip GC0609 is a high-performance, high-power PA (Power Amplifier), which is mainly designed for 800-960MHz ISM band applications such as Wi-SUN, LoRa, automatic meter readers and RFID.

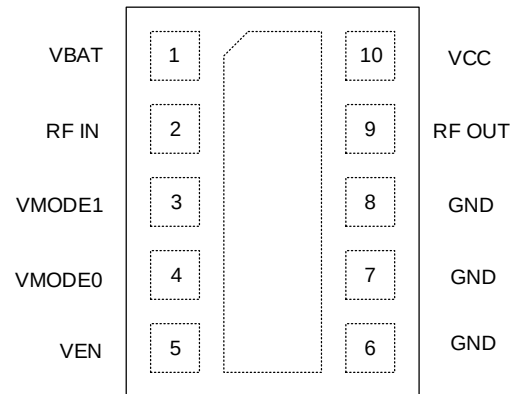
Packaged in a compact LGA 2.5mm x 2mm x 0.75mm, the GC0609 is a 10-pad PA, exhibits high efficiency, low temperature variation, strong reliability and robust ruggedness.



Geo-chip Green™ products are compliant with all applicable legislation and are halogen-free.

Pin Description

Name	Pin	Description
RF IN	2	RF signal input
RF OUT	9	RF signal output
VBAT	1	Power supply for digital and internal drive PA
VCC	10	Power supply for last stage PA
VMODE	3,4	Digital Control signal input for mode selection
VEN	5	PA Enable Control signal
GND	6,7,8	Ground



GC0609 Pinout (Top view)

Absolute Maximum Ratings

Parameters	Symbol	Maximum	Unit
RF Input Power	Pin	5	dBm
Supply Voltage	VBAT	4.5	Volts
Supply Voltage	VCC	5	Volts
Enable Control Voltage	VEN	3.6	Volts
Mode Control Voltage	VMODE	3.6	Volts
Operating Temperature	Tcase	-40~125	°C
Storage Temperature	Tstg	-55~150	°C
ESD (HBM)		+2000	Volts
ESD (CDM)		+750	Volts

Recommended Operating Condition

Parameters	Symbol	Minimum	Nominal	Maximum	Unit
Operating Frequency	f	800	----	960	MHz
Supply Voltage	VBAT	2.1	3.3	3.6	Volts
Supply Voltage	VCC	0.3	4	4.2	Volts
Enable Control Voltage High	VEN_High	1.2	----	3.3	Volts
Enable Control Voltage Low	VEN_Low	0	----	0.3	Volts
Mode Control Voltage High	VMODE_High	1.2	----	3.3	Volts
Mode Control Voltage Low	VMODE_Low	0	----	0.3	Volts
Operating Temperature	Tcase	-40	+25	+105	°C

Note: 1. VEN/VMODE voltage level must not exceed VBAT voltage

2. VBAT must be prior to VEN for the power on sequence.

GC0609 Control Logic

Mode	VEN	VMODE0	VMODE1	VBAT & VCC
High Bias Mode	1	0	0	ON
Low Bias Mode	1	1	1	ON
Low Bias Mode_Temp	1	1	0	ON
Power down mode	0	X	X	ON

"1" denotes high voltage state (> 1.2 V)

"0" denotes low voltage stage (< 0.3 V) at control pins

"X" denotes do not care: either "1" or "0" can be applied

"Low Bias Mode_Temp" The quiescent current and small signal gain is less sensitive to temperature.

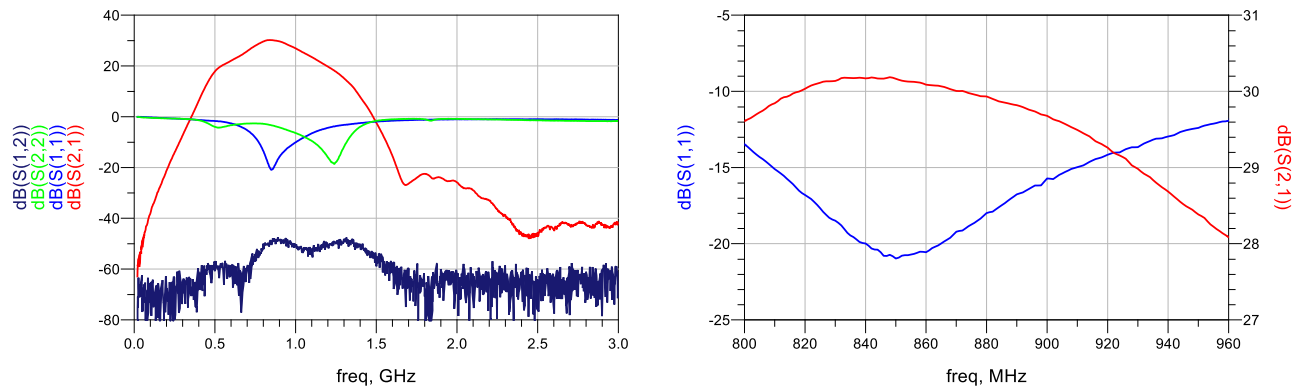
Electrical Specifications (VBAT=3.3V, VCC=4V, Temp=25°C, CW Test)

Parameters	Symbol	Conditions	Min	Typ	Max	Units
Frequency range	f		800	880	960	MHz
High bias mode Gain	IS21I	Small signal Gain @ 880MHz		30.5		dB
Low bias mode Gain				29		dB
Low Bias Mode_Temp Gain				29.5		
Gain variation over frequency	ΔS21I	Small signal		1.5	2	dB
Input return loss	IS11I	Small signal		-12		dB
Output return loss	IS22I	Small signal		-6		dB
Quiescent current	ICQ	High bias mode		68		mA
		Low bias mode		55		mA
		Low Bias Mode_Temp		58		mA
Current consumption @30dBm,880MHz	ICC	High bias mode		550		mA
		Low bias mode		535		mA
		Low Bias Mode_Temp		540		mA
Saturated power	PSAT	High bias mode	31	31.5		dBm
		Low bias mode	31	31.5		dBm
		Low Bias Mode_Temp	31	31.5		dBm
Power added efficiency @Psat, 880MHz	PAE	High bias mode		53		%
		Low bias mode		55		%
		Low Bias Mode_Temp		55		%
Second harmonic	F2	High bias mode @ 30dBm			-10	dBm
		Low bias mode @ 30dBm			-10	
		Low bias Temp @ 30dBm			-10	
Third harmonic	F3	High bias mode @ 30dBm			-15	dBm
		Low bias mode @ 30dBm			-15	
		Low bias Temp @ 30dBm			-15	
Power down mode current					1	uA
DC Enable time*1					3	us
Mode switching time*2					3	us
Ruggedness		Output VSWR = 8:1All phase angles, RF IN = 3 dBm	No damage or permanent performance degradation			
Stability		Output VSWR = 6:1All phase angles				

Typical Performance Curves

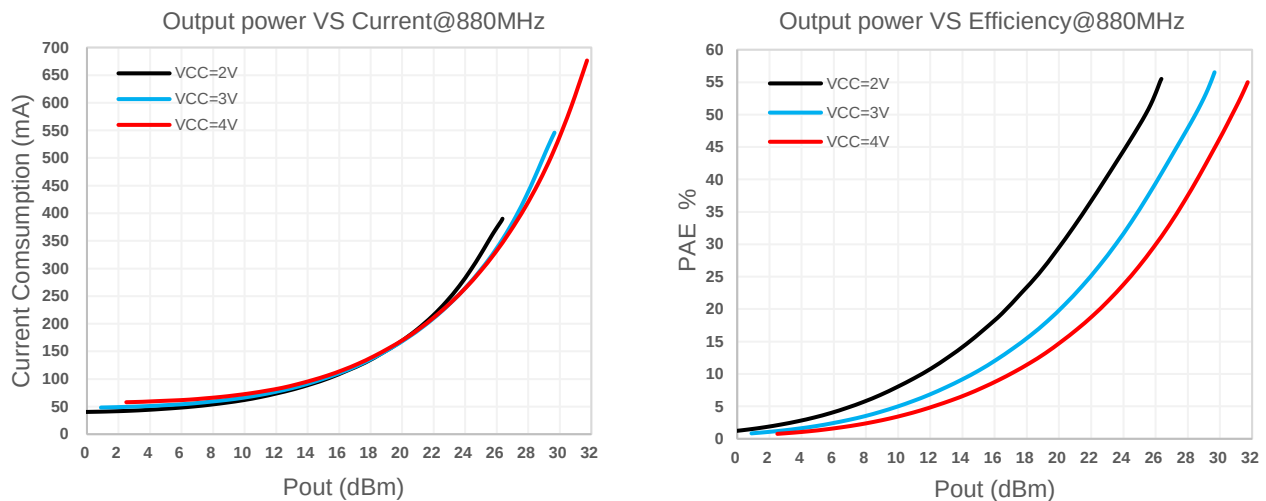
1. S Parameters @ Low Bias Temp Mode

(Conditions: VCC=4.0V, VBAT=VEN=VMODE0=3.3V, VMODE1=0V, Temp=25°C)



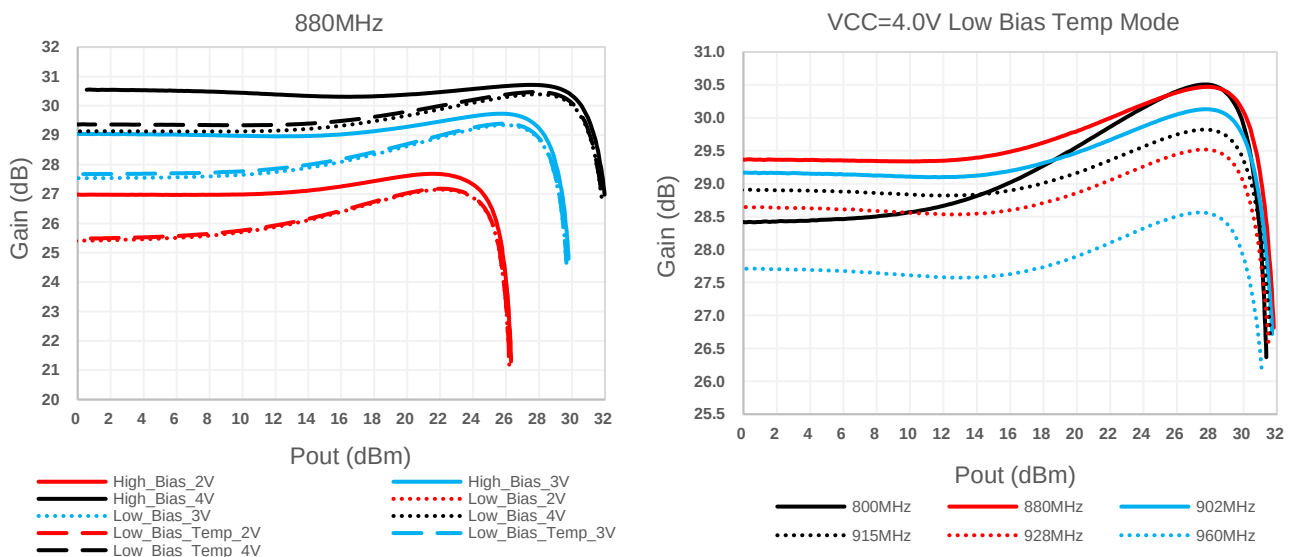
2. Efficiency & Current @ Low Bias Temp Mode

(Conditions: VCC=4.0V, VBAT=VEN=VMODE0=3.3V, VMODE1=0V, Temp=25°C)

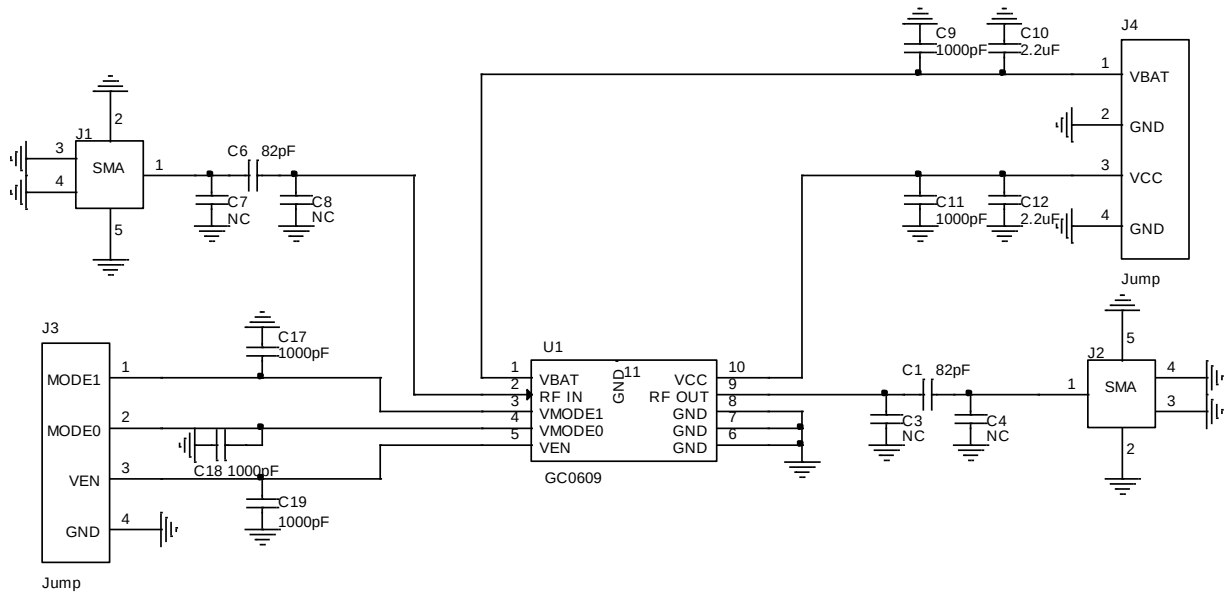


3. Gain vs Pout

(Conditions: VBAT=VEN =3.3V, Temp=25°C)

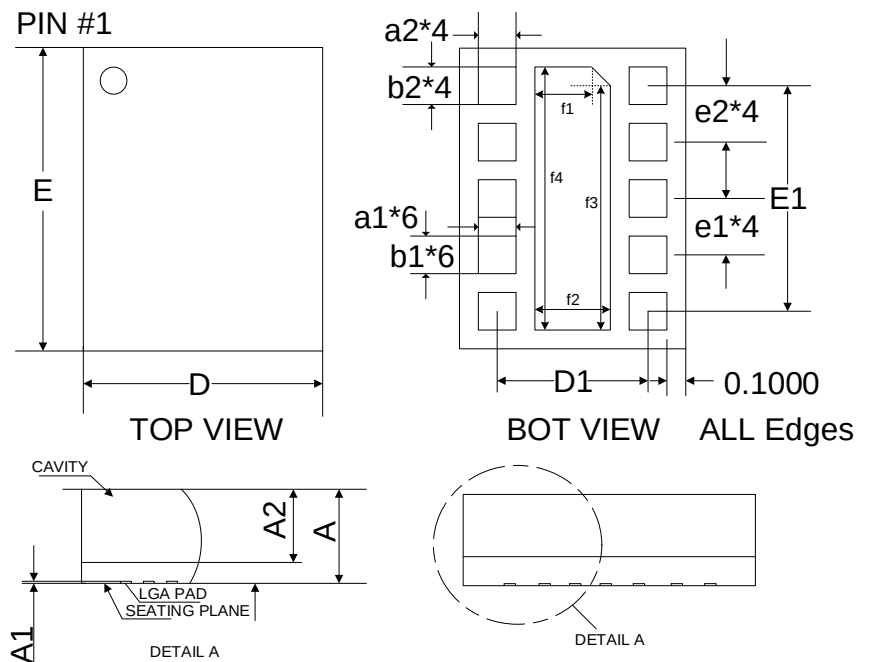


Application Schematic

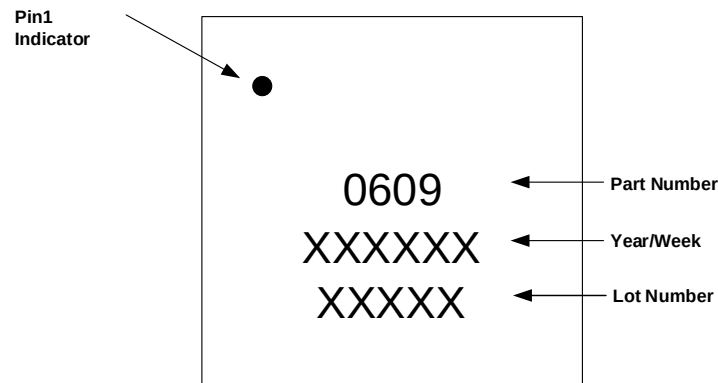


Package Dimensions

Symbol	Dimension in mm		
	MIN	NOM	MAX
A	0.67	0.75	0.81
A1	0.00	---	0.03
A2	---	---	0.78
D	1.90	2.00	2.10
E	2.40	2.50	2.60
D1	---	1.55	---
E1	---	2.031	---
e1	---	0.505	---
e2	---	0.5105	---
a1	0.20	0.25	0.30
b1	0.205	0.255	0.305
a2	0.20	0.25	0.30
b2	0.2175	0.2675	0.3175
f1	---	0.50	---
f2	---	0.70	---
f3	---	2.10	---
f4	---	2.30	---

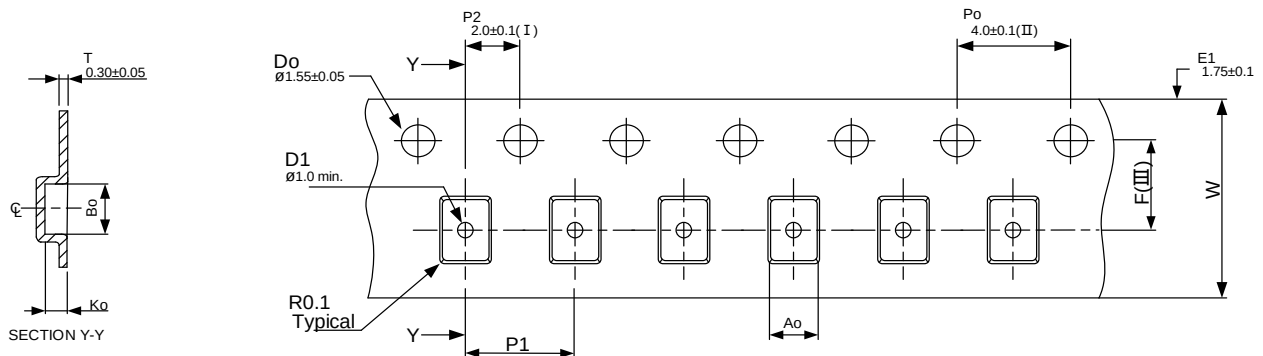


Part Marking



GC0609 Typical Part Markings (Top View)

Packaging Information



A0	2.25 +/-0.05
B0	2.80 +/-0.05
K0	1.10 +/-0.1
F	3.50 +/-0.05
P1	4.00 +/-0.1
W	8.00 +0.3/-0.1

- (I) Measured from centreline of sprocket hole to centreline of pocket.
- (II) Cumulative tolerance of 10 sprocket holes is ± 0.20 .
- (III) Measured from centreline of sprocket hole to centreline of pocket.
- (IV) Other material available.

ALL DIMENSIONS IN MILLIMETRES UNLESS OTHERWISE STATED.

Package Type	Unit Size	Max Reel Diameter	Type Width	Pocket Pitch	Reel Capacity
Tape and Reel	2.5mm x 2mm x 0.75mm	13"	8mm	4mm	3000

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